

PROJECT KNOWLEDGE DOCUMENT

# FinSight AI

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## *AI-Powered Vertical Portfolio Management Platform*

**Version:** 0.1 — Concept Stage

**Date:** March 2026

**Status:** Pre-Development /  
Research

## 1. Project Idea & Vision

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FinSight AI is a vertical AI-powered portfolio management platform designed to act as a personal financial advisor and wealth manager for everyday users. Unlike generic financial apps that display data passively, FinSight AI actively reasons about a user's financial position, monitors global markets and news in real time, and delivers personalized, context-aware suggestions to help users grow wealth, protect savings, and reach financial goals.

### 1.1 The Problem Being Solved

- Retail investors and salaried professionals lack access to the kind of real-time, personalized financial intelligence that institutional investors enjoy.
- Existing apps (Zerodha, Groww, ET Money) are tools — they show data but do not reason about it on behalf of the user.
- Professional wealth managers are expensive and inaccessible to the middle-income segment.
- Global news and geopolitical events affect markets, but most users cannot connect macro events to their personal portfolios.

### 1.2 The Core Vision

*"Give every person — regardless of income or financial literacy — access to an intelligent, always-on financial co-pilot that understands their complete financial picture and helps them make better decisions every day."*

### 1.3 Target Users

- Salaried professionals aged 25–50 with investments in equities, mutual funds, and fixed deposits.
- First-generation investors who lack experience interpreting market movements.
- HNIs (High Net Worth Individuals) who want a second layer of AI-driven insight alongside their human advisors.
- Small business owners managing personal and business finances simultaneously.

## 2. Intention & Strategic Goals

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### 2.1 Short-Term Intention (MVP)

Build a working prototype that demonstrates the core AI reasoning loop: ingest user portfolio → ingest market data → ingest financial news → generate personalized, actionable suggestions.

### 2.2 Medium-Term Intention (V1 Product)

- Launch a consumer-facing web and mobile app with real-time suggestions.
- Integrate with Indian brokers (Zerodha Kite API, Groww, Angel One SmartAPI) and banking APIs.
- Achieve SEBI RIA compliance framework or partner with a licensed advisor.
- Reach 10,000 active users with measurable portfolio improvement metrics.

### 2.3 Long-Term Strategic Goals

- Expand to multi-market coverage: India, US (SEC/FINRA compliant), UAE, and UK.
- Build proprietary financial fine-tuned LLM trained on Indian market data, SEBI filings, earnings calls, and RBI communications.
- Establish a B2B2C channel — white-label the platform for banks, NBFCs, and wealth management firms.
- Become the default AI financial co-pilot for the Indian middle class.

### 2.4 What FinSight AI Is NOT

- Not a trading bot or algorithmic trading system — it advises, it does not execute trades autonomously.
- Not a replacement for SEBI-registered human advisors — it is a decision-support layer.
- Not a generic chatbot — every response is grounded in the user's actual financial data.

## 3. Features (Planned)

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All features below are planned and in design phase. No features are currently live. Features are categorized by priority tier.

### 3.1 Core Features (MVP — Priority 1)

#### 3.1.1 Persistent Portfolio Context

- Secure ingestion of user portfolio data: equity holdings, mutual funds, bank balances, FDs, crypto, real estate.
- Broker and bank API integrations to auto-sync portfolio state (Zerodha Kite, Angel One SmartAPI, Plaid for banking).
- Portfolio is injected into every AI conversation via RAG (Retrieval-Augmented Generation) — the AI always knows where the user stands.
- Manual portfolio entry fallback for users without API-linked accounts.

#### 3.1.2 Real-Time Market Intelligence

- Live market data feeds: NSE/BSE equities, commodity prices, currency rates, global indices.
- Sector-level analysis: the AI understands which sectors the user is exposed to and monitors those specifically.
- Market summary generation: plain-English daily/weekly digest of what happened in the markets and what it means for the user.

#### 3.1.3 Global News Monitoring & Impact Analysis

- Continuous ingestion of financial news (Reuters, Bloomberg, Moneycontrol, Economic Times) and geopolitical news (GDELT, AP, Reuters).
- AI-powered relevance scoring: each news item is classified for sector impact, sentiment, and time horizon.
- Personal impact mapping: the system connects news events to the user's specific holdings — e.g., "Rising crude oil prices could affect your ONGC and HPCL positions."

#### 3.1.4 Personalized Financial Suggestions

- Goal-based advice: suggestions are always framed in context of the user's stated financial goals (retirement, home purchase, child's education, etc.).
- Emergency fund safeguarding: the AI monitors liquidity and warns when savings buffers are under-maintained.
- Rebalancing suggestions: identifies overweight positions and suggests rebalancing toward the user's target allocation.
- Tax optimization hints: flags tax-loss harvesting opportunities, LTCG/STCG implications, and Section 80C utilization.

### 3.2 Enhanced Features (V1 — Priority 2)

- **Risk Profile Engine:** Risk Profile Engine: dynamic risk assessment that adjusts based on market volatility, life events, and user behavior.
- **Goal Simulator:** Goal Simulator: project future wealth scenarios with and without suggested actions using Monte Carlo simulation.
- **Smart Alerts:** Smart Alerts: push notifications for significant events — e.g., a stock in the portfolio drops 5%, RBI rate decision, earnings announcement.
- **Portfolio Stress Testing:** Portfolio Stress Testing: simulate how the portfolio would perform under historical stress events (2008 crisis, COVID crash, 2013 taper tantrum).
- **Competitor Comparison:** Competitor Comparison: compare portfolio performance against relevant benchmarks (Nifty 50, Nifty Midcap, relevant mutual fund category average).

### 3.3 Aspirational Features (V2+ — Priority 3)

- **Voice Interface:** Voice Interface: conversational portfolio queries via voice — "How did my portfolio do this week?"
- **Family Financial Hub:** Family Financial Hub: manage multiple family members' portfolios under one household view.
- **Document Intelligence:** Document Intelligence: upload and analyze SIP statements, ITR filings, insurance policies, and salary slips.
- **Behavioral Coaching:** Behavioral Coaching: detect panic-sell tendencies, overtrading patterns, and provide behavioral nudges grounded in behavioral finance principles.
- **Proprietary Model:** Proprietary Fine-Tuned Model: domain-specific LLM trained on Indian financial corpus for superior market reasoning.

## 4. Technical Architecture

FinSight AI is a vertical AI application — not a new AI model trained from scratch. It is built on top of existing large language models (LLMs) using a combination of RAG (Retrieval-Augmented Generation), agentic orchestration, real-time data pipelines, and narrow ML models for quantitative analysis.

### 4.1 Architecture Overview

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <i>Layer 1: User Interface (Web / Mobile App + Conversational Chat) Layer 2: Orchestration Engine (LangChain / LlamaIndex Agent — decides what data to fetch and how to compose the prompt) Layer 3: Context Assembly (Portfolio DB + Market Feed + News Pipeline — injected via RAG) Layer 4: AI Reasoning Core (LLM API: Claude / GPT-4o / Gemini Pro) Layer 5: Quantitative Models (Narrow ML models for risk scoring, price impact estimation, anomaly detection) Layer 6: Data Infrastructure (PostgreSQL for structured data, Pinecone/Weaviate for vector embeddings, Redis for caching)</i> |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

### 4.2 Key Technical Decisions

| Component     | Technology Choice                    | Rationale                                         |
|---------------|--------------------------------------|---------------------------------------------------|
| LLM Core      | Claude 3.5 / GPT-4o (API)            | Best-in-class reasoning; no training cost         |
| Orchestration | LangChain / LlamaIndex               | Mature agentic frameworks; large community        |
| Vector DB     | Pinecone or Weaviate                 | Fast semantic search over financial documents     |
| Structured DB | PostgreSQL                           | ACID compliance for financial transaction data    |
| Market Data   | Polygon.io / NSE API / Alpha Vantage | Reliable, low-latency market data feeds           |
| News Pipeline | NewsAPI + GDELT + Reuters RSS        | Broad coverage of financial and geopolitical news |
| Backend       | Python (FastAPI)                     | Best ML/AI library ecosystem                      |
| Frontend      | React + React Native                 | Shared codebase for web and mobile                |
| Cloud         | AWS / GCP                            | Scalable infrastructure; good ML tooling          |

### 4.3 AI Approach — What Requires Training vs. What Does Not

- No training from scratch — foundation LLMs are used via API. Training a foundation model costs \$50M–\$500M and is not required.
- Fine-tuning (optional, Phase 2): A Llama 3 70B model fine-tuned on Indian financial data (SEBI filings, earnings calls, RBI policy documents). Estimated cost: \$20,000–\$100,000 in cloud GPU compute.
- Narrow ML models (Phase 1): XGBoost or LSTM models trained on historical market data for quantitative predictions (price impact scores, volatility forecasting, risk scoring). Cheap to train, highly effective.

## 5. Current Progress

|                       |                                       |
|-----------------------|---------------------------------------|
| Overall Project Stage | PRE-DEVELOPMENT — RESEARCH & PLANNING |
| Architecture Design   | COMPLETED — INITIAL DRAFT             |
| Technology Stack      | COMPLETED — SELECTED                  |
| Product Requirements  | IN PROGRESS                           |
| Development           | NOT STARTED                           |
| Regulatory Research   | NOT STARTED                           |
| Team Formation        | NOT STARTED                           |
| Funding               | NOT STARTED                           |

### 5.1 What Has Been Completed

- Concept definition and problem statement finalization.
- High-level architectural design across all 6 system layers.
- Technology stack selection with rationale (LLM API, orchestration, databases, market data sources, cloud provider).
- Feature list definition across 3 priority tiers (MVP, V1, V2+).
- Understanding of AI/ML landscape: distinction between foundation model training, fine-tuning, RAG, and narrow ML models.
- Resource and cost estimation framework established.
- Project knowledge documentation (this document).

### 5.2 Immediate Next Steps

- **Step 1 — PRD:** Finalize detailed Product Requirements Document (PRD) with user stories and acceptance criteria.
- **Step 2 — Regulatory:** Conduct SEBI regulatory research: understand RIA registration requirements; consult a financial regulatory lawyer.
- **Step 3 — PoC:** Build proof-of-concept: connect Claude/GPT-4o API to a static sample portfolio and demonstrate basic reasoning.
- **Step 4 — Team:** Identify and recruit co-founders or key technical hires (backend engineer, ML engineer).
- **Step 5 — Funding:** Define fundraising strategy: bootstrapped MVP vs. pre-seed raise.



## 6. Resource & Cost Estimates

### 6.1 MVP Phase (Months 1–9)

| Resource                          | Estimated Cost (INR)   | Notes                  |
|-----------------------------------|------------------------|------------------------|
| Engineering Team (3 engineers)    | ₹60L – ₹1.2Cr/year     | Backend, ML, Frontend  |
| LLM API Costs (Claude/GPT-4o)     | ₹40K – ₹4L/month       | Scales with users      |
| Market Data Feeds                 | ₹15K – ₹2L/month       | Polygon.io, NSE API    |
| Cloud Infrastructure (AWS/GCP)    | ₹40K – ₹2.5L/month     | DB, compute, pipelines |
| Legal & Regulatory                | ₹10L – ₹20L (one-time) | SEBI RIA consultation  |
| Miscellaneous (tools, design, QA) | ₹5L – ₹10L             | One-time setup costs   |

### 6.2 Optional: Model Fine-Tuning (Phase 2)

| Activity                                  | Cost Estimate        | Timeline            |
|-------------------------------------------|----------------------|---------------------|
| GPU compute for fine-tuning Llama 3 70B   | \$20,000 – \$100,000 | 2–3 months          |
| Training data curation & labeling         | \$10,000 – \$30,000  | 2–3 months parallel |
| ML Engineering (model fine-tuning expert) | ₹30L – ₹60L/year     | Ongoing             |

*Important Note: Fine-tuning is optional for MVP. The initial product can and should be built entirely on top of LLM APIs (Claude, GPT-4o) using RAG and prompt engineering. Fine-tuning becomes relevant only after MVP validation, when domain-specific performance needs to be improved.*

## 7. Regulatory & Risk Considerations

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### 7.1 Regulatory Landscape (India)

- **SEBI RIA:** SEBI RIA (Registered Investment Advisor) license is required to provide personalized investment advice. Fee: ₹5,000 for individuals; minimum qualifications and net worth requirements apply.
- **SEBI RA:** SEBI Research Analyst (RA) registration may be required for publishing market analysis.
- **RBI:** RBI guidelines govern any product that touches banking or payment data — applicable when integrating bank account APIs.
- **DPDP Act:** DPDP Act 2023 (Digital Personal Data Protection Act) applies — user financial data is sensitive personal data and requires explicit consent, data minimization, and right-to-erasure compliance.

### 7.2 Product Design Risk Mitigations

- Frame outputs as "insights, analysis, and education" rather than "personalized investment advice" in early stages to operate without SEBI RIA during MVP.
- Always include prominent disclaimers: "This is not SEBI-registered investment advice. Consult a qualified advisor before making investment decisions."
- Implement a human-in-the-loop layer for high-stakes suggestions in the early version.
- Clearly communicate AI uncertainty — avoid false confidence in suggestions.

### 7.3 Data Security Requirements

- End-to-end encryption for all financial data at rest and in transit (AES-256, TLS 1.3).
- OAuth 2.0 for all broker/bank API integrations — no plain-text credential storage.
- SOC 2 Type II compliance as a target for institutional partnerships.
- Zero data retention policy option for users who delete their accounts.

## 8. Development Roadmap

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| Phase                | Timeline     | Key Deliverables                                                            |
|----------------------|--------------|-----------------------------------------------------------------------------|
| Phase 0 — Foundation | Months 1–2   | PRD finalized, regulatory lawyer engaged, PoC built, team hired             |
| Phase 1 — MVP Build  | Months 3–6   | Portfolio ingestion, LLM reasoning core, basic news pipeline, web app alpha |
| Phase 2 — MVP Launch | Months 7–9   | Broker API integrations, closed beta with 100 users, feedback loops         |
| Phase 3 — V1 Product | Months 10–15 | Smart alerts, goal simulator, stress testing, mobile app, 1000 users        |
| Phase 4 — Scale      | Months 16–24 | Fine-tuned model, B2B2C partnerships, multi-market expansion, 10K+ users    |

## 9. Competitive Landscape

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### 9.1 Existing Players & Gaps

| Competitor                   | What They Do                    | What They Lack                                                           |
|------------------------------|---------------------------------|--------------------------------------------------------------------------|
| Zerodha / Groww              | Stock and MF trading platforms  | No AI reasoning; no personalized advice; data display only               |
| ET Money / Kuvera            | MF investment and goal tracking | No real-time market intelligence; limited AI; no holistic portfolio view |
| Smallcase                    | Thematic portfolio baskets      | Not personalized to individual; no AI advisor layer                      |
| INDmoney                     | Multi-asset aggregation         | Aggregation without intelligent reasoning or personalized guidance       |
| US: Betterment / Wealthfront | Robo-advisors                   | Rule-based, not LLM-powered; limited to the US market                    |

### 9.2 FinSight AI's Differentiation

- Only platform combining: persistent portfolio memory + real-time market data + global news analysis + LLM-powered personalized reasoning.
- AI reasons about the user's specific situation — not generic market commentary.
- Goal-centric framing: every suggestion is tied to the user's stated financial goals and time horizons.
- Built for the Indian market from the ground up — understands Indian instruments (SIPs, PPF, ELSS, SGBs, REITs), taxation, and regulatory context.

## 10. Glossary of Key Terms

| Term                   | Definition                                                                                                                                                        |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LLM                    | Large Language Model — an AI model trained on massive text corpora to understand and generate natural language (e.g., GPT-4o, Claude, Gemini).                    |
| RAG                    | Retrieval-Augmented Generation — technique of injecting external data (portfolio, news) into an LLM's context window at query time, without retraining the model. |
| Fine-tuning            | Retraining a pre-existing model on domain-specific data to improve performance on specialized tasks. Less expensive than training from scratch.                   |
| Vertical AI            | An AI system built for a specific industry domain using existing AI infrastructure, rather than a general-purpose model.                                          |
| Orchestration          | The process of coordinating multiple tools, APIs, and data sources to compose the right context before querying the LLM.                                          |
| Vector Database        | A database optimized for storing and searching high-dimensional embeddings — used for semantic search over financial documents and news.                          |
| SEBI RIA               | Securities and Exchange Board of India — Registered Investment Advisor. A regulatory license required to provide personalized investment advice in India.         |
| DPDP Act               | Digital Personal Data Protection Act, 2023 — India's primary data privacy legislation governing the handling of personal and sensitive digital data.              |
| Monte Carlo Simulation | A computational technique using random sampling to model the probability of different financial outcomes — used in goal projection features.                      |
| LTCG / STCG            | Long-Term Capital Gains / Short-Term Capital Gains — Indian tax categories that determine the rate of tax on investment returns based on holding period.          |